



Introduction to Prompt Engineering

CSIT205 Generative AI – 2025



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*SCHOOL OF COMPUTING AND INFORMATION
TECHNOLOGY*

*FACULTY OF ENGINEERING AND INFORMATION
SCIENCES*

Dr Andrew Yoong, SIM



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Acknowledgement of Country

We acknowledge that Country for Aboriginal peoples is an interconnected set of ancient and sophisticated relationships.

The University of Wollongong (UOW) spreads across many interrelated Aboriginal Countries that are bound by this sacred landscape, and intimate relationship with that landscape since creation. From Sydney to the Southern Highlands, to the South Coast. From fresh water to bitter water to salt. From city to urban to rural.

The University of Wollongong acknowledges the custodianship of the Aboriginal peoples of this place and space that has kept alive the relationships between all living things. The University acknowledges the devastating impact of colonisation on our campuses' footprint and commit ourselves to truth-telling, healing and education.

Journey Artwork by Wodi Wodi
Artist Lauren Henry.

The background is a vibrant rainbow with horizontal bands of red, orange, yellow, green, cyan, blue, and purple. Large, dark blue letters 'U', 'O', and 'W' are positioned on the right side of the image, partially cut off. The 'U' is at the top, the 'O' is in the middle, and the 'W' is at the bottom.

Everyone is welcome here.

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the diversity of our
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Subject introduction



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Subject Description

In this subject, students delve into the world of Generative AI. They explore the foundational principles and practical applications of this cutting-edge field. Through a blend of theory and hands-on experience, students gain insights into how Generative AI empowers artificial intelligence to create original content – ranging from text and images to video and code – based on user prompts. The curriculum covers essential topics, including AI's role in code generation, effective prompt engineering techniques, and the development of advanced generative models like GANs and Transformers. Moreover, students critically examine the ethical implications of Generative AI and its impact on businesses strategy. By the end of the subject, students will have a comprehensive understanding of this promising field and the ability to responsibly apply Generative AI in real-world scenarios.

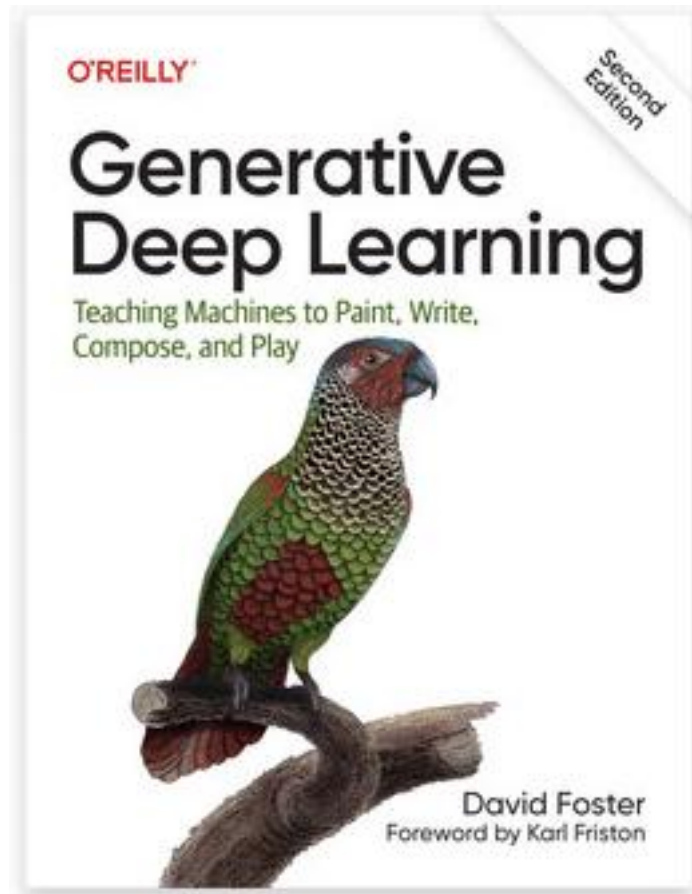
Subject Learning Outcomes (SLOs)

On successful completion of this subject, students will be able to:

1. Demonstrate an understanding of the foundations of Generative AI and its practical applications in text generation, image generation and assisted coding.
2. Apply effective prompt engineering for various tasks.
3. Demonstrate an understanding of techniques to enhance model performance, such as fine-tuning, Retrieval Augmented Generation and Reinforcement Learning using Human Feedback.
4. Identify the impact of Generative AI for industry and employ effective strategies to incorporate Generative AI.
5. Demonstrate ethical and responsible use of Generative AI.

Don't forget 

This is the book we're using



<https://www.oreilly.com/library/view/generative-deep-learning/978109813>

Assessment 2

Assessment 2 – Generative AI business case

- (Day 1 – Form a group of 5 or 6 students and select a responsible student as a group leader. The group leader of a tutorial class is to inform the lecturer of the tutorial class, group members UOW id and names before end of 8th day.)
- (All team members must be present unless there is a valid reason during the interview with CTO during lab 3 and presentation during lab 4.)

Assessment 2

Assessment 2 – Generative AI business case

- File name must be in the form of: TXX_NAME_UOWID.doc where XX is your tutorial group, NAME is the group leader full name (without space or underscore) and UOWID is your 7-digit UOW ID number (not SIM student ID number).
For example, TO2_BobChew_8080426.doc
- (For the report and presentation slides, write the full name of each group member and their UOW ID at the top of first page)

What did we discuss last week?

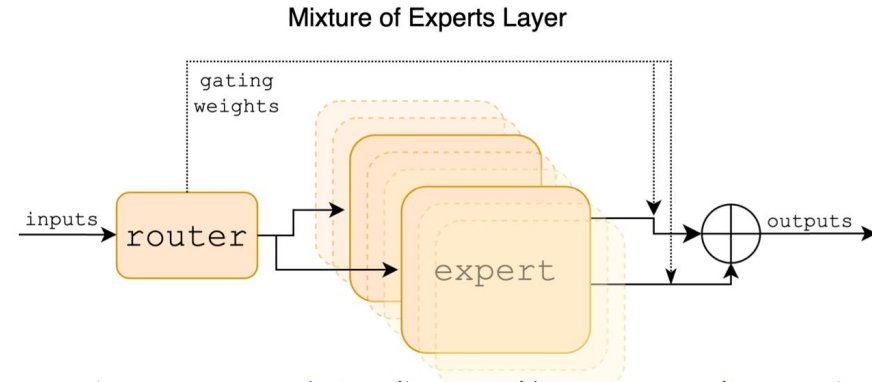


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Mixture of Experts (MoE): do you remember?



- Mixture of Experts (MoE): smaller expert models that each focus on specific sub problems + router that controls experts and output → pre-training with less compute → bigger models + more data!
 - Example: Mixtral 8x7B

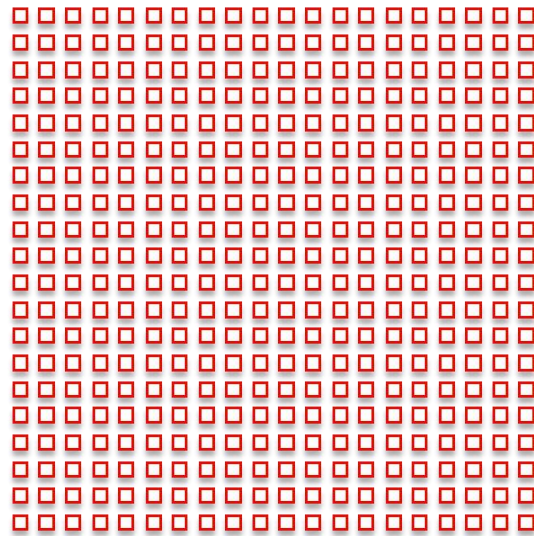


<https://www.analyticsvidhya.com/blog/2024/01/discover-the-groundbreaking-llm-development-of-mixtral-8x7b/>

Small Language Models

- *Lower Parameter Count: fewer parameters compared to LLMs.*
 - *Faster Inference: sub second vs approx. 5+ seconds.*
 - *Cost-Efficient; can run on edge devices*
 - *Task-Specific: may require fine-tuning to increase task-specific performance.*

Llama 3.1 405B (LLM)



Llama 3.2 1B (SLM)



Note

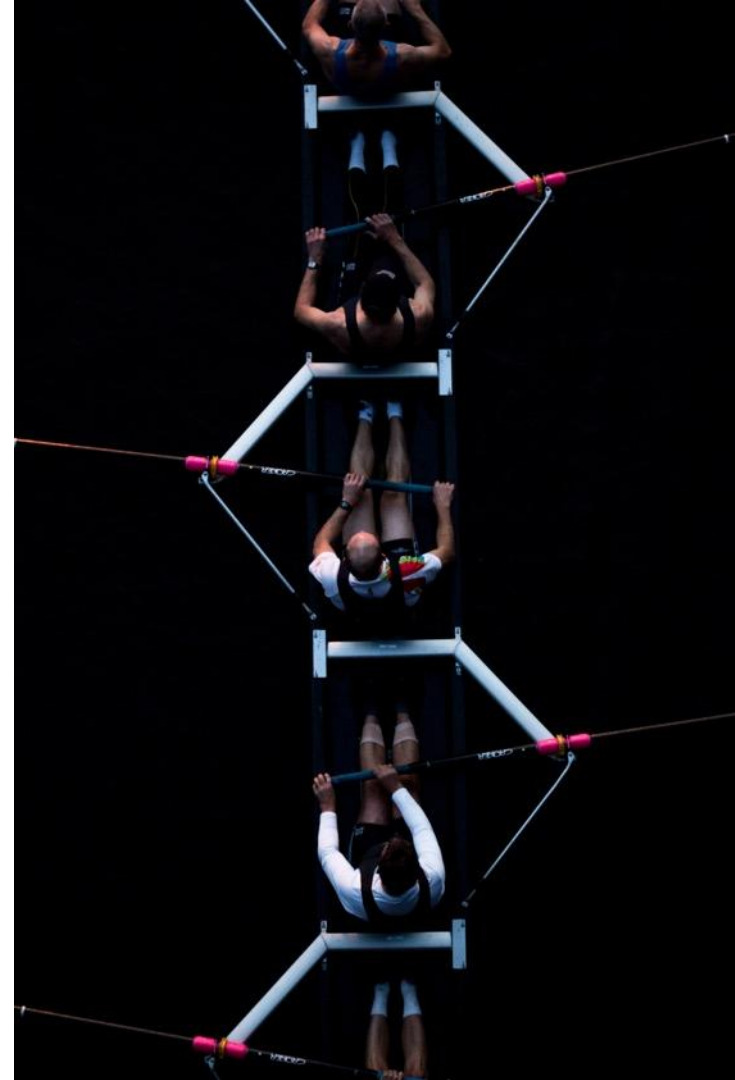
- *Not many have taken Quiz 1 → Quiz 1 will close at the end of week 5.*
 - *This is to prevent competition with Assignment 1 (due week 8).*
 - *Quiz 1 will re-open in week 13.*
- *Last week's Echo360 recording failed.*
 - *Zoom recording has been added to Moodle.*

Why are we here?

CSIT205 Generative AI - Module 2

Prompt engineering

- Prompting 101
- Advanced Prompting



Prompting 101



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The Hot, New High-Paying Career Is An AI Prompt Engineer

By [Jack Kelly](#), Senior Contributor. Jack Kelly covers career growth, job market an...



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Mar 06, 2024 at 12:41pm EST

Prompting introduction

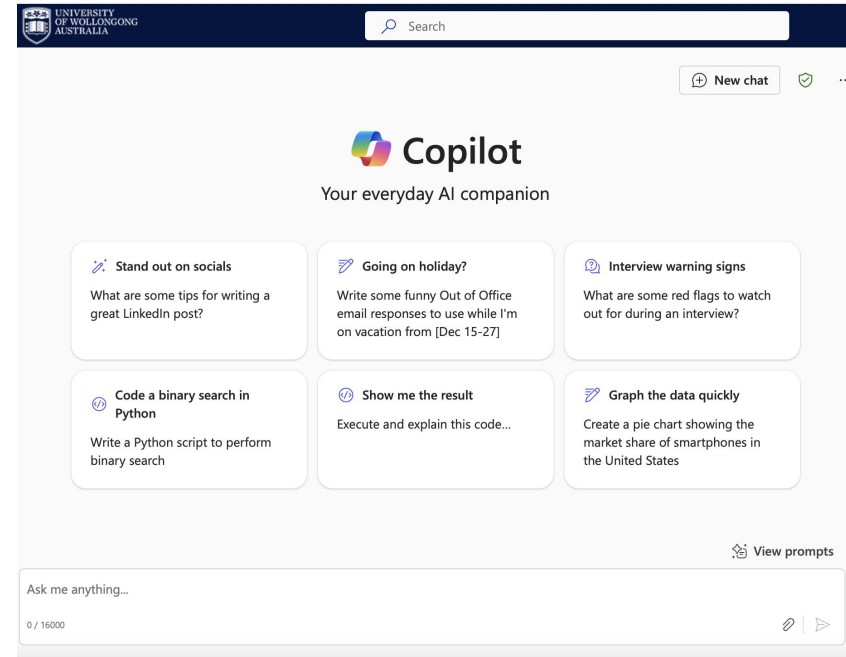


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What is prompting?

Way to interact with a Generative AI model:

- *Instruction: provide a task, like summarisation or image generation.*
- *Input: provide additional context.*
- *Output: provide more details about the output you're after:*
 - *Format*
 - *Tone of voice*

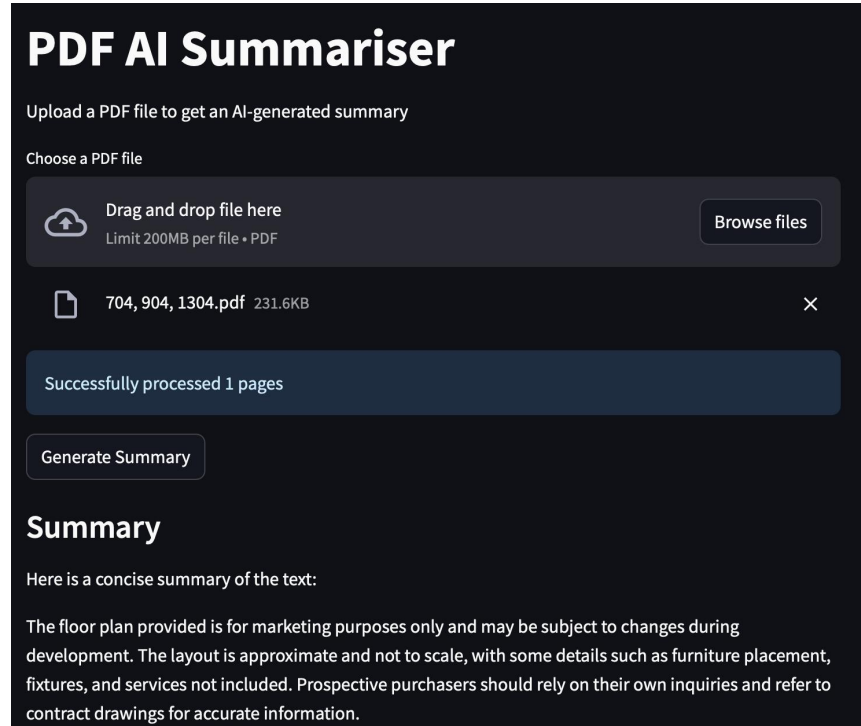


Why do we use prompting?

- *Improved Model Performance: optimise model performance, accuracy, and efficiency.*
- *Increased Interpretability: understand how AI models process and interpret input data. By analysing the output of different prompts we gain insight into the model's strengths, weaknesses, and biases.*
- *Enhanced User Experience: impact user experience in applications like chatbots and virtual assistants. Well-crafted prompts lead to more accurate and helpful responses, improving overall usability.*
- *Reducing Bias and Error: mitigate bias and error in AI models by ensuring that input data is consistent and well-defined. This is essential for applications in healthcare, finance, and law.*
- *Scalability and Efficiency: scale up model performance and reduce computational resources. This enables the deployment of more complex AI models in a wide range of industries and applications.*
- *Improved Explainability: understand AI models by providing insights into how models arrive at their conclusions. This is essential for building trust in AI systems in domains like healthcare and finance.*
- *Reducing Data Overfitting: ensuring that the model generalises well to new, unseen data. This improves model robustness and adaptability in various scenarios.*

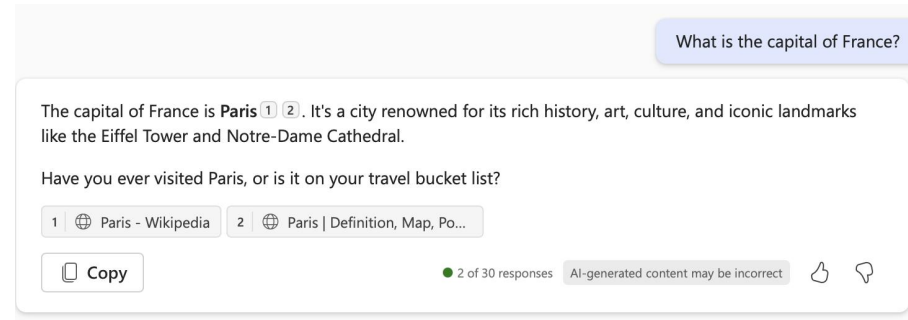
Prompting scope

- *Global: Prompts that influence the entire task or conversation (e.g., "Summarise the main points of this scientific paper")*
- *Task-Specific: Prompts that focus on a specific aspect of the task (e.g., "Explain the concept of LLMs to a 10-year-old")*
- *Local: Prompts that affect a small, localised part of the task (e.g., "Highlight the key differences between Mamba and transformers")*



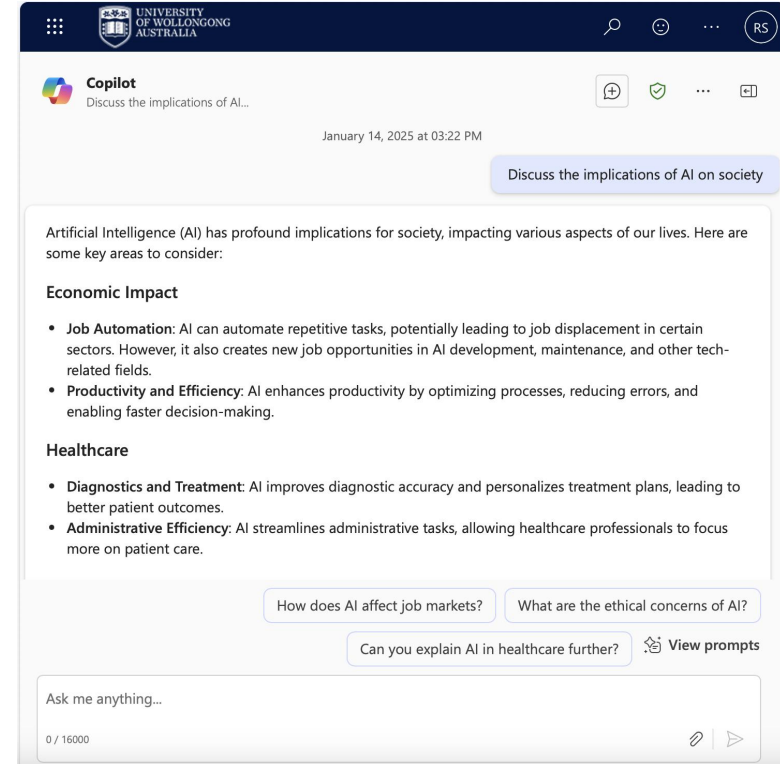
Prompting focus

- *Informational: Prompts that require the model to provide information or answers (e.g., "What is the capital of France?").*
- *Exploratory: Prompts that encourage exploration and discovery (e.g., "Find new uses for this product").*
- *Creative: Prompts that inspire creativity and imagination (e.g., "Write a short story about a character who discovers a hidden world").*



Prompting style

- *Direct: Prompts that are clear, direct, and concise (e.g., "What is the definition of generative AI?").*
- *Indirect: Prompts that require more interpretation or inference (e.g., "The word 'artificial' can be used to describe something that is 'made by human beings' or 'insincere'.")*
- *Open-Ended: Prompts that encourage open-ended discussion or debate (e.g., "Discuss the implications of AI on society").*



Prompting complexity

- *Simple: Prompts that require minimal processing and reasoning (e.g., "What is the colour of the sky?")*
- *Moderate: Prompts that require moderate processing and reasoning (e.g., "Analyse the pros and cons of a new policy")*
- *Complex: Prompts that require advanced processing, reasoning, and understanding (e.g., "Explain the concept of diffusion in image generation models to a non-technical audience")*

```
>>> Explain the concept of diffusion in image generation models to a non-technical audience
**Diffusion in Image Generation Models**

Imagine you're trying to create a beautiful painting from scratch. You start with a blank canvas and some random splatters of color on it. As you paint, you carefully add more layers of color and texture, gradually building up to your desired image.

In image generation models using diffusion, this process is similar. The model starts with a completely random input (like a pixelated noise pattern) and gradually builds up the image through multiple iterations of the following steps:

1. Noise injection: Introduce new random variations into the current output.
2. Diffusion step: Calculate how much noise to inject, based on the patterns learned from previous iterations.
3. Update: Update the model's parameters (weights) based on the new noise and diffusion information.

This process is called a "diffusion step". The model keeps iterating through these steps until it produces an image that looks coherent and realistic.

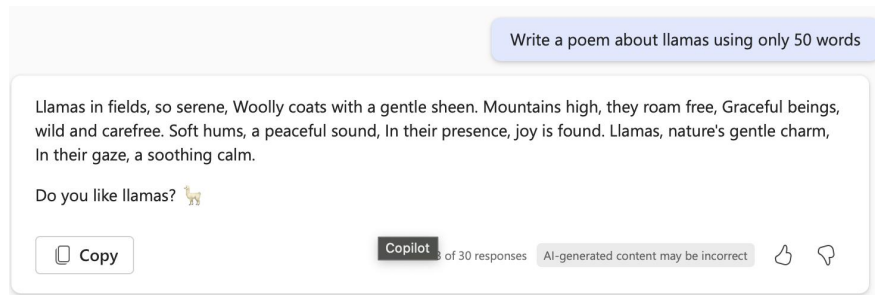
The key advantage of this approach is that the model can generate images in multiple stages, allowing for more control over the final output. It's like adding layers to your painting: each layer builds upon the previous one, creating a rich and detailed texture.

In summary, diffusion in LLMs and image generation models represents a process of gentle nudging, where random noise is introduced to help the model explore different possibilities, gradually building up to coherent and meaningful outputs.
```



Prompting Nuance

- *Strict: Prompts that require strict adherence to specific rules or constraints (e.g., "Write a poem about llamas using only 50 words")*
- *Flexible: Prompts that allow for some flexibility or interpretation (e.g., "Describe the main character's motivations in a novel")*
- *Ambiguous: Prompts that are intentionally vague or open-ended, requiring the model to make assumptions or inferences (e.g., "Write a story about a character who discovers a hidden truth")*

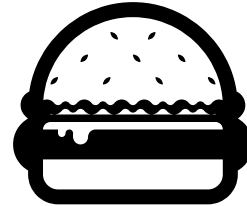


10-minute Break!

This will help you learn



10 minutes



Fill out the following blanks for Prompting Style



- <BLANK>: Prompts that require more interpretation or inference (e.g., "The word 'artificial' can be used to describe something that is 'made by human beings' or 'insincere'.")
- <BLANK>: Prompts that are clear, direct, and concise (e.g., "What is the definition of generative AI?")
- <BLANK>: Prompts that encourage open-ended discussion or debate (e.g., "Discuss the implications of AI on society")

A. open-ended, indirect, direct

B. Indirect, open-ended, direct

C. Indirect, direct, open-ended

D. Direct, indirect, open-ended



What is not a reason to use prompting?

- A. Retraining an LLM
- B. Reducing Bias and Error
- C. Improved Explainability
- D. Enhanced User Experience

"Discuss the implications of AI on society" is an example of:



- A. A simple prompt.
- B. A moderately complex prompt.
- C. A complex prompt.
- D. An invalid prompt.

Basic Prompting Techniques



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Zero-shot prompting (Wei et al, 2022)

- *Instruction*
- *No example*
- *Fully relies on the LLM's pre-existing knowledge and skills*

describe this code: `def inner_product(a, b): result = 0 for i, j in zip(a, b): result += i * j return result`

Sure! Here's a detailed description of the code:

- Function Definition:** `def inner_product(a, b):`
 - This line defines a function named `inner_product` that takes two parameters, `a` and `b`.
- Initialize Result:** `result = 0`
 - This line initializes a variable `result` to 0. This variable will be used to accumulate the sum of the products of corresponding elements from `a` and `b`.
- For Loop with Zip:** `for i, j in zip(a, b):`
 - The `zip(a, b)` function pairs elements from `a` and `b` together. The `for` loop iterates over these pairs, with `i` taking values from `a` and `j` taking corresponding values from `b`.
- Accumulate Product:** `result += i * j`
 - Inside the loop, the product of `i` and `j` is calculated and added to `result`.
- Return Result:** `return result`
 - After the loop completes, the function returns the accumulated `result`, which is the inner product (or dot product) of the two input lists `a` and `b`.

Here's the complete code for reference:

```
def inner_product(a, b):  
    result = 0  
    for i, j in zip(a, b):  
        result += i * j  
    return result
```

This function effectively computes the inner product of two lists by multiplying corresponding elements and summing the results. Let me know if you have any questions or need further clarification!

One-shot prompting

- *Instruction*
- *One example*
- *The LLM 'learns' from one example.*
 - *This may make it more challenging to generalise to other tasks or domains.*

Task: classify the following movie review into 'positive', 'negative', or 'sarcastic'.

Input: "Masterpiece of cinema to view if you've viewed all other movies on Netflix"

Output: sarcastic

Input: "The best two hours of my day."

Output:

Output: positive

This review expresses genuine enjoyment and satisfaction, indicating that the movie provided a highly enjoyable experience for the reviewer.

Copy

Copilot

5 of 30 responses

AI-generated content may be incorrect



Few-shot prompting

- Instruction
- Few example
- Allows the LLM to 'learn' in-depth about how a) input is processed and b) to produce / format output.

There is an optimal number of examples per model, but performance commonly maxes out at around 5 examples.

Task: classify the following movie review into 'positive', 'negative', or 'sarcastic'.

Input: "Armageddon got some astronomy right. For example, there is an asteroid in the movie, and asteroids do indeed exist."

Output: sarcastic

Input: "Masterpiece of cinema to view if you've viewed all other movies on Netflix"

Output: sarcastic

Input: "Best two hours of my day"

Output: positive

Input: "Two hours of my life I'll never get back."

Output: negative

Input: "Unforgettable experience"

Output:

Output: positive

The review "Unforgettable experience" suggests a highly memorable and likely enjoyable time, indicating a positive opinion.

Copy

9 of 30 responses

AI-generated content may be incorrect

Copilot

Advanced Prompting



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Chain-of-Thought (CoT) prompting (Wei *et al.*, 2022)

Adds reasoning steps to the prompt

🏆 Achieved state-of-art performance on Math Word Problems (GSM8K) in 2022.

Four main attributes (Wei *et al.*, 2022):

- “allows models to decompose multi-step problems into intermediate steps”.
- “provides an interpretable window into the behavior of the model”.
- Can be used for math problems, commonsense reasoning and symbolic manipulation.
- Can be applied as part of few-shot prompting.

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Prompt Chaining

- Break complex tasks up into sub tasks.
- Use the output of one prompt as input for the next prompt:
 - Prompt 1: Get all menu categories and format them using <category></category> tags
 - Output 1: Soups, appetizers, salads, sandwiches, fried seafood, broiled fresh seafood, from the land, house specialties, dessert.
 - Prompt 2: pick a combination of starter, main and dessert.
 - Output 2: Lobster bisque, fried shrimp, ask your server for daily selection.

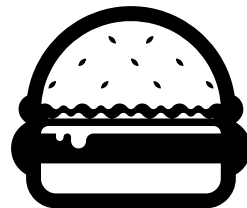


6-minute Break!

This will help you learn



6 minutes



Assignment: zero- and one-shot prompting



15 minutes

1. Go to your favourite Generative AI (note: Copilot (<http://copilot.cloud.microsoft/>) is free and secure for UOW students and staff).
2. Grab a text from Medium, Daily.dev, or AnalyticsVidhya.com, e.g. <https://www.analyticsvidhya.com/blog/2025/01/scikit-llm/>.
3. Apply zero-shot prompting, e.g. to summarise the article.
4. Apply one-shot prompting to control the output format, e.g. number of words, sections, or tone of voice.
5. Make a screenshot and send it to rorys@uow.edu.au, so we can share it with the rest of the class.



15 minutes

Assignment: prompt chaining

1. Go to your favourite Generative AI (note: Copilot (<http://copilot.cloud.microsoft/>) is free and secure for UOW students and staff).
2. Grab a text from Medium, Daily.dev, or AnalyticsVidhya.com, e.g. <https://www.analyticsvidhya.com/blog/2025/01/scikit-llm/> .
3. Get the key concepts of the article.
4. For each of the key concepts, try to find a scientific reference article.

Evaluating and testing prompts



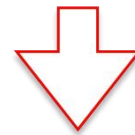
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Metrics for evaluating prompts

- *Perplexity*: A measure of how well the model predicts the next token in the sequence.
- *BLEU score* (Papineni et al., 2002): A metric for evaluating translation quality.
- *ROUGE score* (Lin, 2004): A metric for evaluating question-answering performance.
- *Human evaluation*: Using human evaluators to assess output quality.

$$p_n = \frac{\sum_{C \in \{Candidates\}} \sum_{n\text{-gram} \in C} Count_{clip}(n\text{-gram})}{\sum_{C' \in \{Candidates\}} \sum_{n\text{-gram}' \in C'} Count(n\text{-gram}')}.$$

² $Count_{clip} = \min(Count, Max_Ref_Count)$. In other words, one truncates each word's count, if necessary, to not exceed the largest count observed in any single reference for that word.



Candidate: the the the the the the the.

Reference 1: The cat is on the mat.

Reference 2: There is a cat on the mat.

$$Count(n\text{-gram}') = Count(Candidate) = 7$$

$$Count_{clip}(the) = \min(7, 2) = 2$$

$$p_n = 2/7$$

Types of Prompt Evaluation

- *Relevance: Your prompt should be relevant to the task you're trying to accomplish. If you're trying to generate content for a specific topic, your prompt should relate to that topic.*
- *Clarity: Your prompt should be clear and easy to understand. If your prompt is confusing, your model is likely to generate confusing output.*
- *Specificity: Your prompt should be specific enough to guide your model toward the intended output. Vague prompts can lead to irrelevant or off-topic output.*

Types of Output Evaluation

- *Accuracy: How well does the model perform on the task?*
- *Relevance: Does the output answer the question or complete the task?*
- *Coherence: Is the output clear, logical, and easy to understand?*
- *Contextual understanding: Can the model capture nuances and subtleties in the prompt?*

Gender Shades



Testing Prompts for Bias and Fairness

- *Bias testing: Identifying biases in the model's response, such as stereotypes or discriminatory language*
- *Fairness metrics: Using metrics to evaluate fairness, such as equal opportunity and non-discrimination tests*
- *Adversarial prompt engineering: Testing prompts designed to mislead or deceive the model*

Adversarial prompt attacks (Zhu et al., 2023)

- *Character-level: manipulate texts by introducing typos or errors to words, e.g., by adding, deleting, repeating, replacing, and permuting characters for certain words.*
- *Word-level: replace words with synonyms or contextually similar words to deceive LLMs.*
- *Sentence-level: add irrelevant words to the end of a prompt to distract LLMs. For instance, append `and true is true, and false is not true, or and true is true to the end of a prompt.`*
- *Semantic-level: simulate linguistic behaviour non-native English speakers, translate their prompts to English to introduce linguistic nuances and variations that could potentially impact LLMs.*

Original	Insert	Delete	Swap	Sub-C	Sub-W
foolish awfully cliches	f oolish awfull y clich es	folish awfully clichs	fooilsh awfluly clcihes	fo0lish awfully cliches	silly terribly cliche

TextBugger (Li et al., 2019)

Wrap-up



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Summary

- *Basic prompting techniques include Zero-shot, one-shot, and Few-shot prompting.*
- *We can extend prompts using Chain-of-Thought (CoT) reasoning.*
- *We have benchmarks like BLEU and ROUGE to evaluate LLMs, but we should not forget human evaluation.*
- *You can potentially break LLMs by using adversarial attacks.*

For next lecture

- *Retrieval Augmented Generation (RAG)*

References and Resources

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Questions



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